
ISyE 6740 – Computational Statistics – Fall 2024

Project Group 088 - Project Proposal

Team Name: Term Project Group # 88

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Project Title: People Analytics - Predicting Employee Attrition with Data-Driven Approach

Problem Statement

Employee attrition is an important analysis for the HR department as high employee turnover will bring challenges and make negative impacts, such as reducing productivity, increasing overall cost, decreasing morale, disrupting team dynamics, and finally impairing long-term sustainability.

Suppose we are data scientists in a company and are collaborating with the HR department to build employee attrition prediction models and help them to establish data-driven employee retention strategies and reduce overall costs associated with recruitment and training. We will use an HR dataset containing 34 predictor variables, including factors such as age, education, distance from home, job satisfaction, salary, marital status, and others. The target variable for this analysis is attrition, categorized as either "Yes" (employee has left) or "No" (employee has stayed).

The goal of this project is to predict if employees are at risk or tendency of resignation based on their working situation and employee survey, and identify patterns of employee turnover. Additionally, we aim to identify key factors that may drive employee resignations, which will help uncover underlying issues within the company and provide insights of HR benchmarks into the competitive landscape. This study will help the organization to better understand their employees, implement effective people policies and HR interventions, and finally provide a healthier working environment and positive organizational culture.

Data Source

The IBM HR Analytics Employee Attrition & Performance dataset, sourced from Kaggle, is a dataset created by IBM data scientists for HR analytics purposes, this dataset is modeled closely on real-world HR data, making it highly representative of actual organizational environments. It consists of **1,470 rows** and **35 columns**, with each row representing an individual employee, representing employee data from an anonymous company. The dataset's main objective is to uncover the factors leading to employee attrition (whether employees leave the company or not). We will leverage feature selection techniques to make the dataset more efficient and better.

Key attributes in the dataset include:

- Employee demographic information: such as age, gender, marital status and education
- Job-related attributes: including job role, department, monthly income, job satisfaction, environment satisfaction, and years at company.
- Performance metrics: performance rating, training times last year, and years since last promotion.
- Attrition: the target variable, indicating whether the employee has left the company or stayed.

This dataset provides a comprehensive view of both employee profiles and organizational attributes, making it suitable for predictive modeling and analysis of employee turnover patterns. The wide variety of predictor variables enables in-depth analysis using machine learning and statistical methods to gain insights into the factors influencing attribution.

Methodology

- **Exploratory Data Analysis (EDA) and Preprocessing**

We will begin by conducting a detailed exploratory data analysis (EDA) to understand the structure of the IBM HR Analytics dataset and uncover initial patterns. This will involve:

- Descriptive statistics to summarize key variables.
- Visualizations such as bar plots and histograms to illustrate relationships between features.
- Correlation analysis to assess the linear relationships between numerical variables and attrition.

- **Data Cleaning and Preparation**

Next, we will clean the dataset to ensure its quality for machine learning:

- Handling missing data: Use imputation methods where necessary.
- Encoding categorical variables: Convert certain features into numeric form using techniques such as one-hot encoding or label encoding.
- Standardizing features: Scale continuous features (e.g., monthly income, age) to standardize the dataset, which is essential for algorithms like SVM and KNN.
- Splitting the dataset: divide data into training and testing sets for model evaluation.

- **Initial Statistical Analysis (Regression)**

As an initial groundwork step, we will run a logistic regression model to determine whether the 34 predictor variables have statistically significant effects on employee attrition. This step will also serve as a baseline for comparison with more complex machine learning models.

- **Supervised Classification Models**

We will then implement a series of supervised classification models to predict employee attrition. Each model will be tuned and evaluated based on accuracy, precision, recall, and F1 score to determine which performs best in predicting employee turnover:

- Baseline model: **Logistic Regression**.
- **Logistic Regression with Regularization (L1/L2)**: To prevent overfitting and manage multicollinearity among the features.
- **K-Nearest Neighbors (KNN)**: A simple yet powerful method, particularly useful in detecting clusters of high-attrition employees.
- **Support Vector Machines (SVM)**: Effective for classification in high-dimensional spaces.
- **Naive Bayes**: A probabilistic model that assumes independence between predictors.
- **Decision Trees**: To capture non-linear relationships between variables.
- **Random Forest**: An ensemble of decision trees helps improve accuracy, reduce variance.
- **Gradient Boosting & XGBoost**: Powerful boosting methods to further improve predictive performance by iteratively focusing on errors made by previous models.
- **AdaBoost**: Another boosting algorithm that enhances model performance by combining multiple weak classifiers into a strong classifier, particularly effective in reducing bias.
- **Neural Networks**: To capture more complex patterns within the dataset.
- **Ensemble Methods**: We'll combine models using techniques such as stacking and bagging to leverage the strengths of each algorithm, improving overall predictive performance.

- **Dimensionality Reduction Techniques**

To further improve model efficiency and interpretability, we may apply dimensionality reduction techniques such as:

- Principal Component Analysis (PCA): To reduce the dimensionality of the dataset by identifying the most influential factors.
- Truncated SVD: To compress the feature space, especially for algorithms sensitive to high-dimensional data (e.g., SVM).

- **Supplementary Datasets and Feature Engineering**

In addition to the primary dataset, we will explore the integration of supplementary datasets:

- Demographic factors: If possible, we may consider incorporating race and citizenship to enrich our analysis. This can reveal if foreign employees have lower attrition rates due to visa constraints or if employees earning below-market salaries are more likely to leave.
- External salary benchmarks: Compare industry salary averages by job role, examining whether employees earning below market standards exhibit higher attrition rates.
- Additional factors: If possible, we could include more qualitative factors such as work culture or engagement metrics to enhance our model.

- **Innovation**

- **Model Comparison:** By comparing a wide range of models, we aim to capture the most important patterns in employee attrition.
- **Regularization:** Logistic regression with L1 and L2 regularization helps prevent overfitting, making the models more robust.
- **Ensemble Methods:** Combining strengths from various models to boost predictive performance and uncover hidden relationships in the data.
- **Comprehensive Feature Engineering:** we will explore external factors like salary benchmarks or visa status, which can offer deeper insights into the drivers of attrition.
- **Dimensionality Reduction:** Incorporating techniques like PCA and Truncated SVD to streamline the feature space and enhance model performance.

Evaluation Strategies

To evaluate the effectiveness of our models, we will employ standard classification metrics such as **accuracy**, **precision**, **recall**, **F1 score**, and **AUC-ROC**. These metrics will provide a comprehensive assessment of each model's performance in predicting employee attrition. Accuracy will measure the overall correctness of the model's predictions, while precision and recall will provide insights into how well the model is identifying actual cases of attrition versus false positives. The F1 score will offer a balance between precision and recall, and the AUC-ROC curve will give a deeper understanding of model performance across different thresholds.

Additionally, we will apply **cross-validation techniques** such as K-fold and Monte Carlo cross-validation to ensure model robustness and generalizability. This will help avoid overfitting and ensure that the models perform consistently across different subsets of the data. To further enhance performance, we will use **hyperparameter optimization** methods such as grid search or random search to fine-tune model parameters, ensuring the best possible performance for each algorithm.

We will visualize the results through confusion matrices, which will clearly show the number of true positives, true negatives, false positives, and false negatives. This will help us compare the performance of the various models and select the most suitable one for predicting employee attrition.

Furthermore, **dimensionality reduction techniques** like PCA (Principal Component Analysis) will be considered to reduce the complexity of the data and improve computational efficiency. We will also conduct **feature selection** to identify the most important predictors of attrition, ensuring that the model focuses on the most relevant factors.

By integrating these evaluation methods, we will be able to make a well-informed decision about the best-performing model. The final model will not only offer high predictive accuracy but also align with the HR department's practical needs, providing actionable insights into employee retention strategies. The HR team can use this predictive tool to identify employees at high risk of attrition and implement targeted interventions, ultimately improving employee retention and optimizing resource allocation.

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